HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and developed **AI-ready data models** in Snowflake, integrating enterprise and investment data sources to support AI-driven portfolio management tools, ensuring scalability and reuse across multiple applications.
- Implemented data transformations and ETL/ELT processes using **SQL and Python** to connect diverse enterprise data sources with curated datasets for consumption by AI applications, processing approximately 15,000 records per hour.
- Embedded **data quality, governance, and lineage** tracking into data solutions, ensuring AI applications access trusted, controlled, and traceable data, reducing data ambiguity by an estimated **20%**.
- Collaborated with AI engineers and investment stakeholders to translate complex business and application needs into practical data requirements and scalable data solutions, including RAG and embeddings patterns.
- Supported modern AI data patterns such as **RAG, embeddings, and vector search**, integrating structured and unstructured data for AI applications and agent-based solutions.
- Worked closely with the Data Strategy team on their **Snowflake implementation** to identify, shape, and make required data readily available for consumption by AI-developed portfolio management tools.
- Containerised data pipeline services using **Docker** and deployed them to an **AWS EKS** cluster, configuring basic resource limits for predictable rollouts and improved operational stability.

SKILLS

Data Modelling & Integration – Snowflake, Data Transformation, ETL, ELT, Data Pipelines, SQL, Python, Data Warehousing, Data Integration

AI & Data Enablement – AI Data Patterns, RAG, Embeddings, Vector Search, Structured Data, Unstructured Data, AI-Assisted Development

Data Quality & Governance – Data Quality, Data Governance, Data Lineage, Trusted Data, Controlled Data, Traceable Data

Collaboration & Stakeholder Management – AI Engineers, Investment Stakeholders, Data Strategy, Architects, Business Requirements, Application Needs

Cloud & Infrastructure – AWS, Snowflake, Docker, Kubernetes, EKS

Programming & Scripting – Python, SQL, Bash Scripting

PROJECTS

AI Data Foundation for Portfolio Analytics

- Designed and implemented a curatated data layer in **Snowflake**, integrating financial transaction and market data for use in AI-driven portfolio risk analysis.
- Developed **Python** scripts to automate the extraction, transformation, and loading of data from multiple enterprise sources, supporting an estimated **30,000** daily data points.
- Integrated a **vector search** capability for unstructured research reports, enabling AI agents to perform semantic searches against the investment data repository.

Enterprise Data Integration for Investment Tools

- Built reusable **data integration patterns** connecting core enterprise data warehouses with front-office portfolio management tooling, improving data accessibility for analysts.
- Developed **SQL**-based data transformations and maintained data lineage documentation for key investment datasets, ensuring compliance and auditability for regulatory reporting.
- Collaborated with a cross-functional team of data engineers and portfolio managers to define data requirements for a new analytics platform.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)